



CAPSTONE PROJECT PROPOSAL AND TEAM CONTRACT

Operation Catch the Hacker

A red versus blue security exercise. Where good defense meets a little sparkle.

TEAM	Melanie Seguin (Mel), Jan Lelie, Kenny Panyavong, Mugolie Queen
COURSE	Fullstack Academy Cybersecurity Capstone
STAGE	Step 3 of 9, Proposal and Team Contract
DATE	Prepared June 2026

The Mission

Unicorns Wireless Corp, our made-up scenario company, runs a small practice network for training. Leadership wants to know how well the company would hold up if someone tried to break in, so they approved an internal security exercise. Our team runs it from both sides. The Red Team plays the attacker and tries to get into a practice machine. The Blue Team watches, catches the attack, and locks the system down. Everything happens inside an isolated lab, on machines we own, with permission. No real customer data and no real systems are ever touched.

What We Will Deliver

By the end of the capstone our team will show a full attack, detection, and defense cycle on a practice target, plus a short phishing awareness study. Our goals:

- Show a complete attack, catch, and lock-down on a practice machine.
- Prove we can catch an attack using the tools we learned in class.
- Study how attackers now use AI to write phishing, and make a simple guide to spot it.
- Work as one team across red and blue, even with a member joining remotely.

Scope

IN SCOPE

- Our own isolated lab VMs
- An authorized attack on our practice target
- Traffic capture, detection rules, firewall hardening
- An AI-assisted phishing awareness study
- Written findings and a recorded demo

OUT OF SCOPE

- Any real network, company, or person
- Any machine we do not own
- Collecting real personal data
- Anything outside the lab
- If it is not our practice machine, we do not touch it

Team Roles and Responsibilities

Four people, four different jobs, split across a Red Team and a Blue Team. Each role is its own piece of the project so the work is shared evenly.

Red Team Operator

RED TEAM · OFFENSE

Melanie Seguin (Mel) Works independently, records own segment

- Set up a practice target and map it with Nmap to find its open services.
- Guess a weak SSH login with a brute force tool, and capture the whole attack with Wireshark.
- Post the capture file and clear screenshots to the team Slack for the Blue Team.
- Record a short walkthrough of her steps so the team can present this part while she is away.

Recon and Social Engineering

RED TEAM · OFFENSE

Jan Lelie Works ahead or syncs, then records

- Gather open information about the practice company the way an attacker would.
- Use the AI assistant to draft sample phishing emails for the awareness study.
- Note the warning signs in each sample so they can be studied.
- Hand the phishing material to the Blue Team for the defense guide.

Detection Analyst

BLUE TEAM · DEFENSE

Kenny Panyavong Part of the live recorded session

- Open the Red Team capture file and find the attack inside it with Wireshark.
- Write Snort rules that catch the scan and the break-in.
- Use the AI assistant to explain the alerts in plain English and write the findings.
- Present the detection part in the live demo.

Mugolie Queen Part of the live recorded session

- Turn on the firewall, block the attacker, and close the weak service.
- Re-run the scan to prove the attack is now blocked, with a before and after.
- Turn Jan's phishing samples into a one-page guide on spotting phishing.
- Present the defense part in the live demo.

Mel runs her part on her own and shares the saved capture file, so the rest of the team is never stuck waiting. The other three can build their parts live together and record it, which is part of the assignment, or work ahead on their own and record afterward. Jan can switch to the Blue Team if the group prefers, as long as all four roles stay covered.

How We Work Together (Team Agreement)

We agree to the following while we build this project. These are the rules we hold each other to.

1. We make decisions as a group and aim for agreement. If we disagree, we talk it through and pick what is best for the project.
2. We listen to each other. We repeat back what we heard, ask questions, and clarify before we react.
3. We keep communication clear and short. Updates and all files go in our team Slack channel, and written pieces are saved as PDFs.
4. We give and accept feedback kindly, and we use it to make the work better.
5. We meet on a set schedule to sync. The live build sessions are recorded as part of the assignment.
6. Mel works remotely. She finishes early, shares a recorded walkthrough and a short runbook, and stays reachable by message.
7. Everyone hits their deadlines, or speaks up early if something slips. We cover for each other when we need to.
8. We keep the work honest and inside the lab. We only test machines we own, with permission.

Tools and Lab Setup

- **VMs:** Parallels on the MacBooks (Mel and Jan), or VirtualBox on the Lenovo and the class Windows machines.
- **Attacker and target:** two Kali Linux machines. Kali already includes Nmap, Hydra, Wireshark, and Snort, so there is nothing extra to install.
- **The attack:** an Nmap scan plus an SSH password guess with Hydra against a target that has a weak login.
- **Defense:** the UFW firewall built into Linux.
- **AI assistant:** used to explain alerts and to write phishing samples to study.
- **Sharing:** all files and write-ups are posted in the team Slack channel. Written pieces are saved as PDFs.

Good news on hardware: because both the attacker and the target are Kali, the whole project runs on the MacBooks in Parallels, on the Lenovo, and on the class Windows machines. Every tool is already inside Kali, so there is nothing new to download.

Milestones (Following the Capstone Process)

Our project maps to the nine steps in the capstone process. Here is what each stage means for us and who leads it.

Capstone Stage	What it means for our project	Who leads
3. Proposal and Team Contract	This document. We agree on the plan and all sign it.	All four
4. Proof of Concept	One small test that proves the idea works: Mel runs one attack and posts the capture, Kenny gets one Snort alert to fire.	Mel and Kenny
5. Minimum Viable Project	The full attack, detect, and defend cycle working once, rough but complete end to end.	All four
6. Complete Project	Polished version with the phishing guide, clean screenshots, and written findings.	All four
7. Demonstration and Feedback	Live demo to the class. Blue Team drives, Red Team supports.	Blue leads
8. Video Due	Recorded walkthrough of the whole project. Mel's segment is pre-recorded.	All four
9. Graduation Showcase	Present and share with family and friends.	All four

Signatures

By signing below, we agree to the roles, scope, and team agreement in this document.

Team Member	Role	Signature	Date
Melanie Seguin	Red Team Operator	_____	_____
Jan Lelie	Recon and Social Eng.	_____	_____
Kenny Panyavong	Detection Analyst	_____	_____
Mugolie Queen	Defense and Hardening	_____	_____